American Universities: The cost and outcomes of a College Education

1. **Section 1. Introduction** – The goal of this project is to present data that shows key metrics involving higher education, ones that may drive decision making for prospective students. Students need to balance tuition, possible borrowing costs, and expected outcomes. They may also need to assess admissions probability to see what is a good fit for them.

Link to Tableau Public Interactive Graphs and Dashboard: Gernhardt_Final_Project

2. **Section 2. Data Description**

3. **Describe the domain of the data set:** The rows in the data file represent individual institutions across the United States. The data I am focusing on is about 4-year colleges and universities, some of which offer higher degrees.

4. **Describe the data file:** The file only includes relevant metrics collected by the US Department of Education. The field names and data types are listed below for reference:

Field	Data Type	Description
College ID	String	Numeric value associated with each university, used as a primary key
College Name	String	Name of Institution
City	City	City of Institution
State	State	State of Institution
Highest Degree Offered	Category	3- Bachelor's Degree, 4-Graduate Degree. Other degree types were filtered out for this analysis
Institution Type	Category	1- Public, 2- Private nonprofit, 3-Private for-profit
Admission Rate	Percentage	Rate of applicant acceptance
Undergraduate Enrollment	Numeric	Total number of undergraduate students
Average Annual Cost of Attendance	Dollars	Most recent annual cost of attendance
In-State Tuition and Fees	Dollars	Tuition cost for in state students
Out-of-State Tuition and Fees	Dollars	Tuition cost for out of state students
Graduation Rate	Percentage	Percentage of students who graduated taking 150% of expected time. For undergraduate students, this means within 6 years of enrollment.
Share Receiving Federal Student Loans	Percentage	Percentage of students using federal loans to fund their education
Median Federal Student Loan Debt - Graduates	Dollars	Median Student loan debt at graduation
Median Earnings 6 Years After Entry	Dollars	Median Salary starting 6 years after start of enrollment
Median Earnings 10 Years After Entry	Dollars	Median Salary starting 10 years after start of enrollment

5. **Clearly state the number of rows and columns:** There are 16 columns and 1,777 rows in this data set.
6. **Describe the data source:** The data source is the U.S. Department of Education's College Scorecard, who pulls this data to make sure families can make informed decisions about choosing universities by having information on tuition, admissions, and expected outcomes. It can also be used by institutions themselves as well as policymakers. This data is the most recent data available and was updated in the summer of 2026.
7. **Section 3. Data Cleaning Strategies** – this data took a bit of cleaning before it could be used in Tableau. The raw data had an excess of columns that were not used for this project. Additionally, many rows have null values, but they were inconsistent- some numerical fields had inconsistent ways of expressing blank values. For consistency, I made these all "NA" values so they can easily be excluded from the data when appropriate.

   Additionally, I needed to use the Data Dictionary provided by my source, because I needed to understand the column headings that were often abbreviated. I then used these plain English translations for my column headers. This dictionary also provided additional context for what exactly the data represents. This data cleansing, in my experience, tends to be the most time-consuming part of any visualization project.
8. **Section 4. Clean Dataset** – The clean data set has been simplified to only display metrics about each university that are relevant to the goals below or that are helpful additions that can be used if further analysis is desired. A sample of the data is below.

College	College Name	City	State	Highest Degree Offered	Institution Ty	Admission Ra	Undergraduate Enrollment	In-State Tuition and Fees	Out-of-State Tuition and Fee	Graduation Rate	Median Federal Student Loan Debt - Graduates	Median Earnings 6 Years After Entry	Median Earnings 10 Years After Entry
100654	Alabama A & M University	Normal	AL	4	1	57.95%	6,124	$10,024	$18,634	24.03%	$31,000	$27,851	$39,059
100663	University of Alabama at Birmingham	Birmingham	AL	4	1	88.18%	11,635	$9,098	$22,562	64.23%	$22,300	$46,572	$50,907
100706	University of Alabama in Huntsville	Huntsville	AL	4	1	68.57%	6,591	$12,132	$26,408	64.29%	$20,705	$55,610	$56,901
100724	Alabama State University	Montgomery	AL	4	1	97.55%	3,477	$11,248	$19,576	30.34%	$31,000	$27,453	$34,018
100751	The University of Alabama	Tuscaloosa	AL	4	1	76.65%	33,227	$12,180	$34,172	73.37%	$22,750	$52,233	$56,118
100830	Auburn University at Montgomery	Montgomery	AL	4	1	91.57%	2,542	$9,700	$20,668	32.60%	$25,000	$38,216	$43,743
100858	Auburn University	Auburn	AL	4	1	45.92%	26,816	$12,890	$34,922	81.95%	$21,000	$54,629	$61,042
101189	Faulkner University	Montgomery	AL	4	2	73.07%	1,549	$23,920	$23,920	37.40%	$23,000	$37,313	$41,523
101365	Herzing University-Birmingham	Birmingham	AL	3	2	93.26%	442	$13,450	$13,450	12.50%	$21,500	$35,295	$34,238
101435	Huntingdon College	Montgomery	AL	4	2	69.49%	879	$29,070	$29,070	47.98%	$27,000	$42,147	$47,328
101480	Jacksonville State University	Jacksonville	AL	4	1	77.69%	7,036	$12,894	$23,334	54.72%	$22,189	$40,158	$41,504
101587	University of West Alabama	Livingston	AL	4	1	42.65%	2,974	$10,990	$20,090	34.70%	$24,944	$37,106	$38,617
101693	University of Mobile	Mobile	AL	4	2	77.89%	1,307	$26,910	$26,910	56.85%	$26,500	$37,411	$43,640
101709	University of Montevallo	Montevallo	AL	4	1	53.85%	2,151	$13,948	$26,968	54.10%	$24,000	$35,406	$40,439
101879	University of North Alabama	Florence	AL	4	1	87.31%	5,811	$12,120	$22,320	54.91%	$22,077	$38,295	$42,601
101912	Oakwood University	Huntsville	AL	4	2	45.22%	1,153	$22,512	$22,512	44.62%	$27,000	$30,969	$41,219
102049	Samford University	Birmingham	AL	4	2	82.36%	4,206	$40,150	$40,150	77.04%	$19,500	$49,466	$54,742
102094	University of South Alabama	Mobile	AL	4	1	70.96%	8,879	$10,116	$19,092	52.67%	$24,929	$41,307	$45,018
102234	Spring Hill College	Mobile	AL	4	2	76.57%	782	$24,054	$24,054	52.10%	$27,000	$41,358	$51,166
102270	Stillman College	Tuscaloosa	AL	3	2	62.25%	706	$12,126	$12,126	25.00%	$29,067	$29,790	$34,087
102298	Talladega College	Talladega	AL	4	2	85.42%	701	$15,650	$15,650	30.91%	$28,500	$22,613	$30,032
102368	Troy University	Troy	AL	4	1	96.04%	9,805	$10,176	$20,352	50.30%	$25,000	$39,392	$41,691
102377	Tuskegee University	Tuskegee	AL	4	2	48.68%	2,630	$25,386	$25,386	54.98%	$27,000	$38,332	$43,878
102632	University of Alaska Southeast	Juneau	AK	4	1	63.29%	907	$7,056	$20,640	28.05%	$19,111	$38,811	$41,500
102669	Alaska Pacific University	Anchorage	AK	4	2	96.45%	457	$20,760	$20,760	47.22%	$23,500	$47,803	$47,105
104151	Arizona State University Campus Immersion	Tempe	AZ	4	1	89.89%	64,674	$12,223	$33,139	68.04%	$19,500	$52,843	$58,967

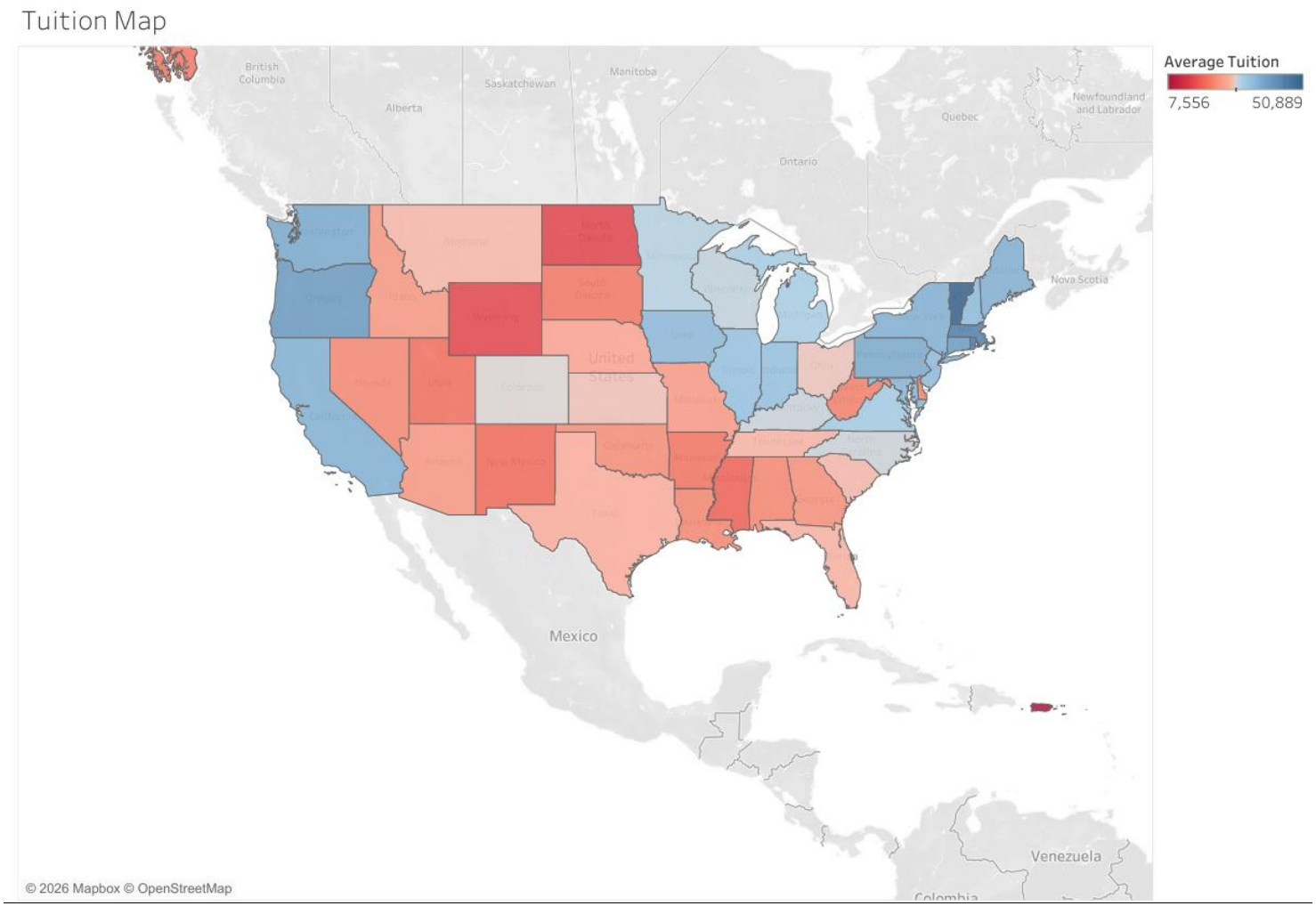
9. **Section 5. Visualization Tools** – I utilized Tableau to visualize this project. Tableau has the versatility to show maps and dashboards, which are effective data storytelling tools. The presentation is also often cleaner than excel, especially when it comes to filtering and labeling. It also allows drilling down into specific data points much easier.

Link to relevant data files: Gernhardt_Final_Project

10. **Section 6. Visualizations and Stories**

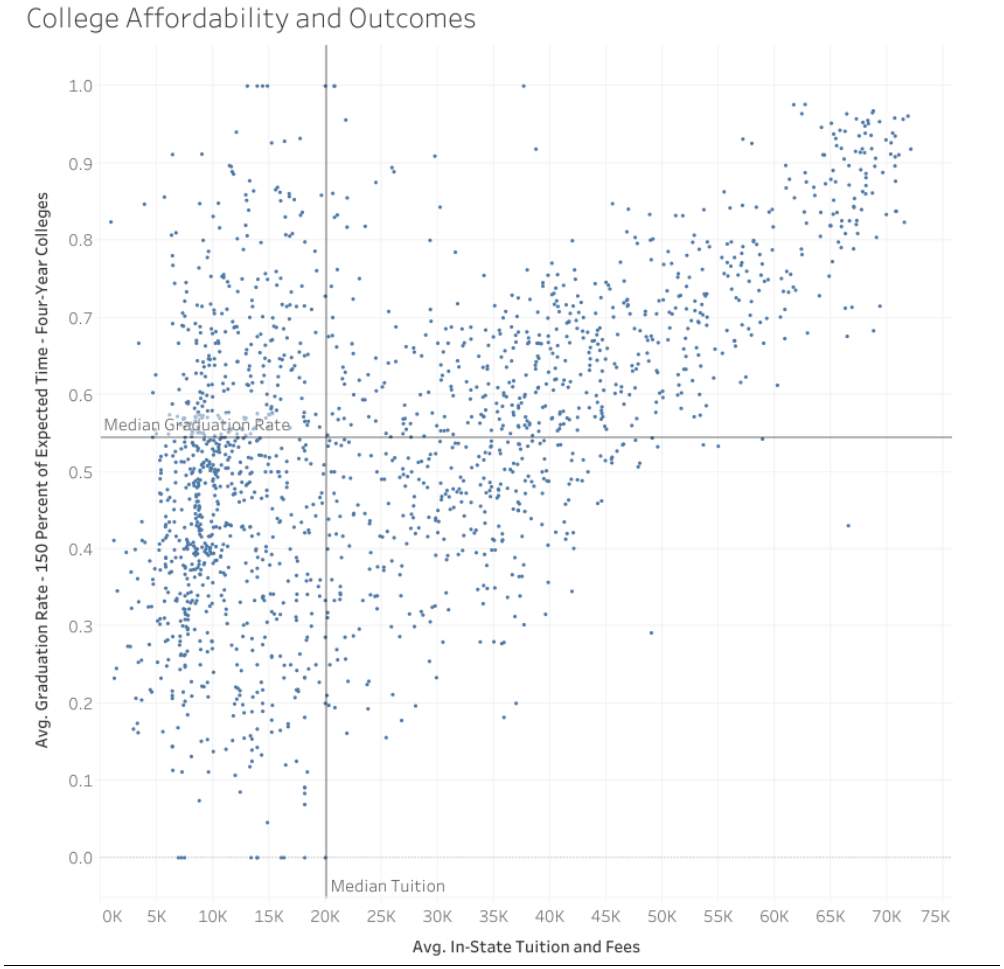
11. **Use Dashboards & storyboards to represent your charts and findings.**

<u>Goal 1:</u> Understand what states have higher or lower tuition rates and be able to filter this by in-state and out-of-state rates.



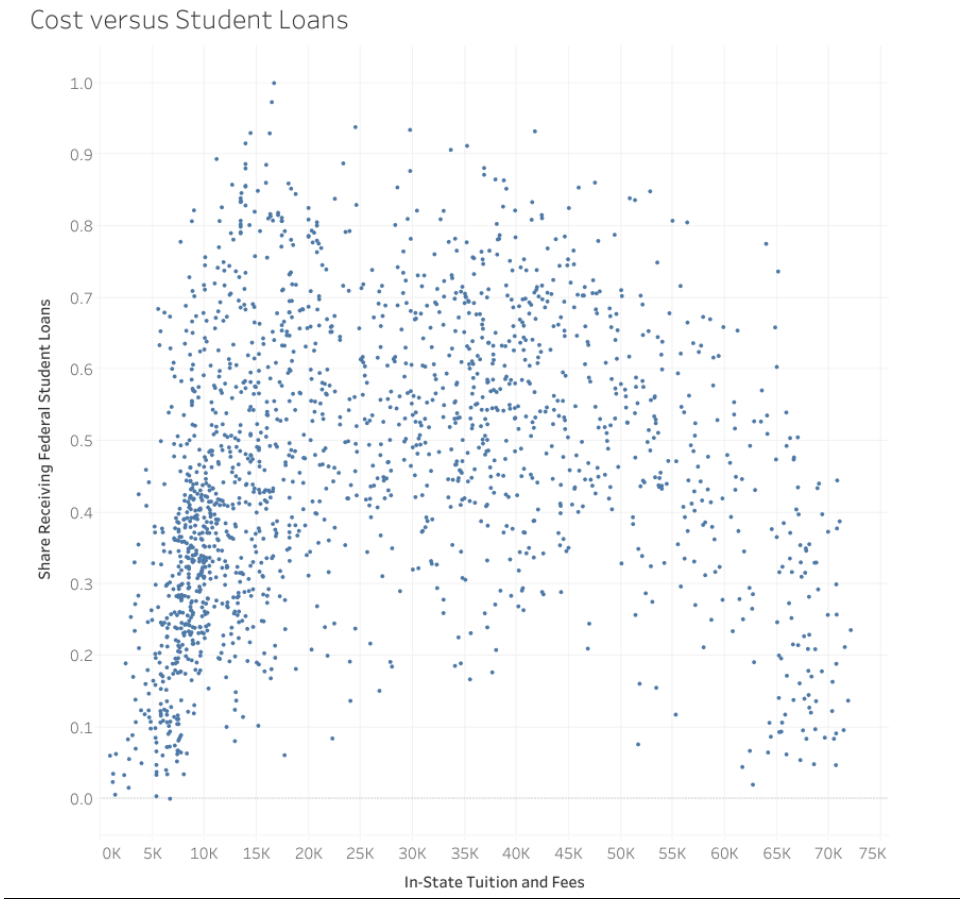
Story: This map gives us a general view of tuition rates by region for out-of-state tuition. As we can see, some of the more expensive options are on the coasts, as well as some midwestern states. Vermont is the most expensive, averaging $50 thousand dollars a year. Another notable feature is that of North Dakota and Wyoming- who have average tuitions of $14 thousand and $15 thousand, respectively. These are likely influenced by factors like cost of living, and is influenced by more prestigious and expensive universities driving up the average that largely exist in costal areas.

<u>Goal 2:</u> To determine the correlation between tuition price versus expected graduation rates



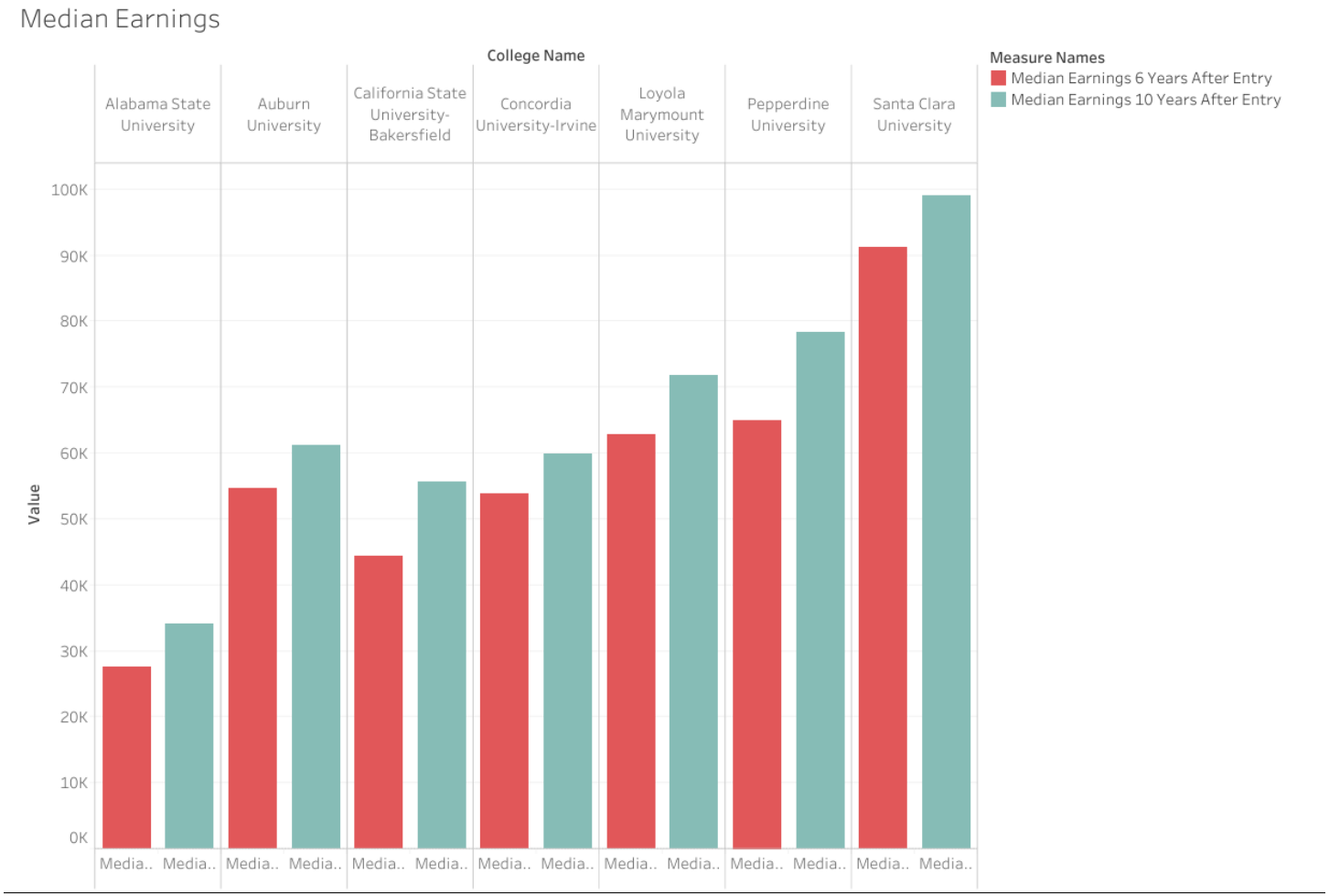
Story: An interesting trend appears when analyzing average tuition and graduation rate. This relationship is almost linear, but there are wildly differing outcomes for universities with lower tuition costs. However, when looking at very expensive universities, there is a clear trend that appears where graduation rates increase significantly. This could be caused by the selection of students that take on high tuition rates, as there is a chance they are likely motivated to complete their education after some significant financial investments.

<u>Goal 3:</u> To determine the relationship between tuition costs and the percentage of students who take out federal loans.



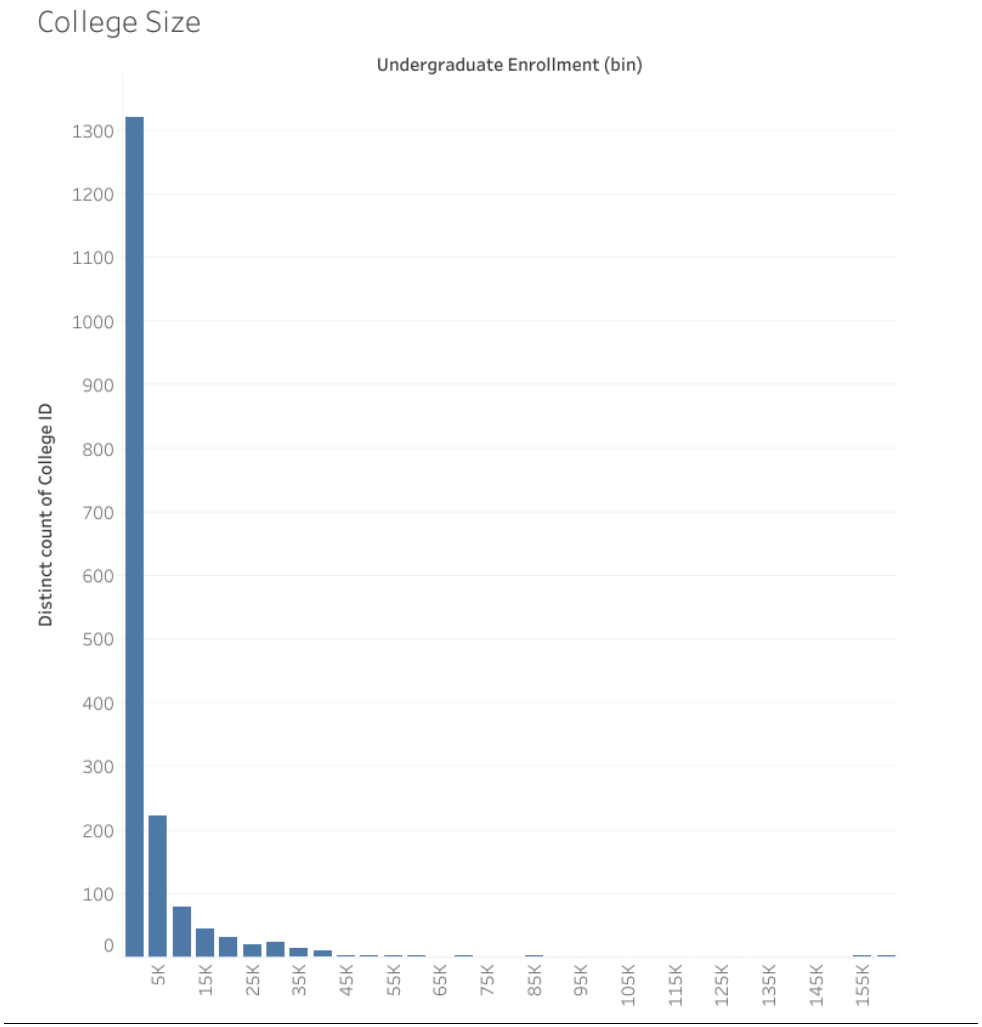
Story: This trend is particularly revealing. Low-cost universities tend to have less students taking out loans, which seems unsurprising. The more fascinating trend is that very expensive universities actually see a decreased number, on average, of students that take out federal loans. This may relate to how students select universities. Those with the means to afford an expensive education are more likely to go to them, as students who need to take loans out will opt for more financially viable options. Moderately priced universities seem to hit a sweet spot where the cost is low enough for them to be willing to take out the loans, but too high for them to be afforded outright.

<u>Goal 4:</u> To look at individual universities and see the median earnings at various points, including 6 and 10 years after entry to the university.



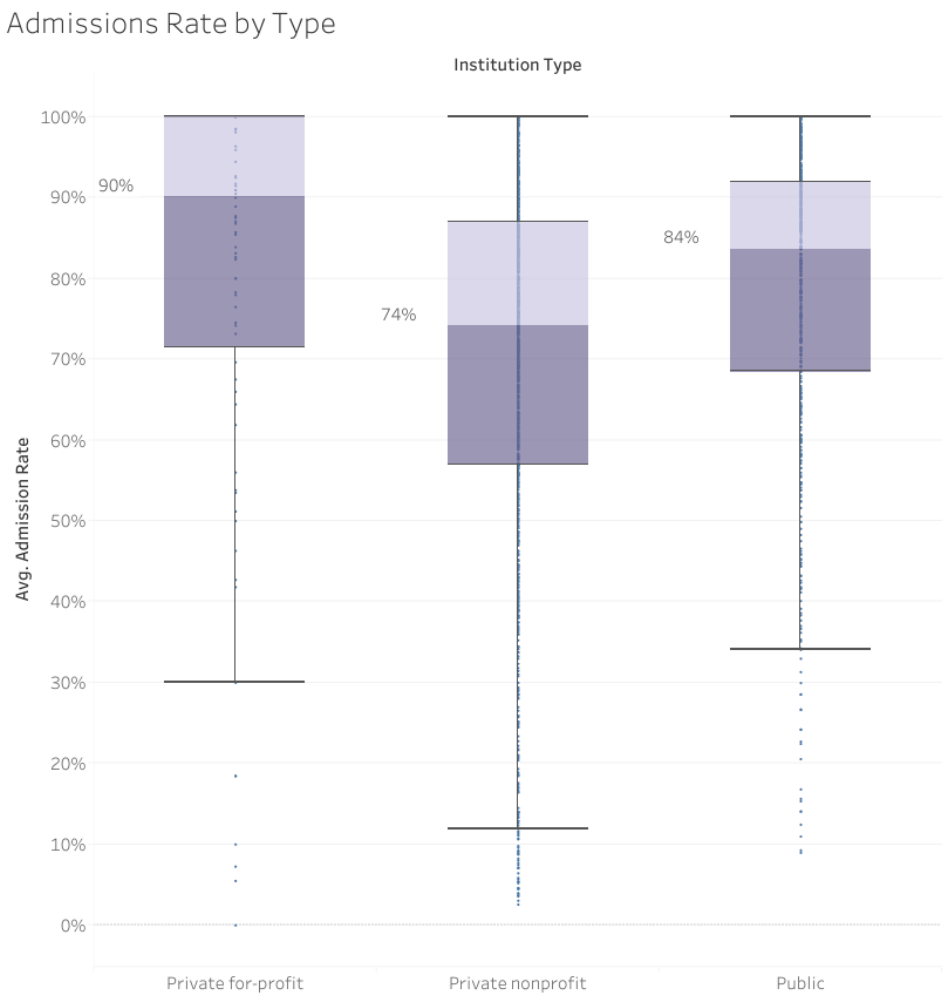
Story: This is a very effective tool when looking at a particular university a student may want to attend. Understanding expected income after graduation can help manage expectations of prospective students, and their ability to pay off student loans quickly. One example is Santa Clara University- its location in Silicon Valley likely contributes to its high earning potential of around 91 thousand just six years after enrollment. Though it can depend on field of study and plans after graduation, these are good factors to consider when applying.

Goal 5: To understand the typical university size and see how common certain university sizes are.



Story: A histogram showing college sizes gives us some interesting insight. Many students look to large, established institutions as options for their continued study. However, in terms of choices, around 74% of colleges have less than five thousand students. These universities are often overlooked and may provide more options to potential students worth exploring. They may have a unique focus, and more opportunities for involvement and faculty contact.

Goal 6: **To understand admissions rates based on the University type- distinguishing from Public, Private nonprofit, and private for-profit universities.**



Story: This shows acceptance rate among various types of universities, as well as how spread these values are. Private, for profit institutions seem to have considerably higher acceptance rates. This is likely due to the natural profit motive- more students means more income for the university. On the other hand, private institutions are harder to get into, on average. The advantage of this graph is that it still shows that despite institution type, acceptance rates can vary significantly. For example, although the median admissions rate of public universities is 84%, only half of them lie between 69% and 92%, an already very wide range.

12. Section 7. Conclusions:

There are quite a few conclusions we can make from these visualizations. In terms of tuition costs, colleges vary widely. Geographically, there are certain regions that have lower costs on average. Some of the notable states that average on the higher end is Vermont at $50 thousand, Massachusetts at $45 thousand, and California at $37 thousand. These states have some of the most sought-after universities in the country, and that can sometimes come with high tuition costs, as well as high cost of living.

Diving deeper into graduation rates, low-cost universities have wildly variable outcomes when it comes to graduation rates. A notable outlier is Columbia University in New York, which has tuition of over $70 thousand dollars, but also has a 96% graduation rate. Many universities like this are highly selective and likely attract highly motivated students that complete their degrees. On the other end, some lower cost universities may attract some students that are tentative about completing higher education, because any sunken cost that may result will be less impactful. On the other end, some of these cheaper universities may attract motivated students that are cost conscious. This may warrant individual case studies to see if they explain this general trend.

The most fascinating trend revealed from these graphs is the peak in student loan borrowers being middle-cost universities. Their moderate cost means there is still need for student loans, but high-cost universities may deter those same students without the means to pay. University of Chicago, for example, has a very high tuition of $70 thousand, but is extremely selective and has less than one percent of students taking on federal loans. This data, as well as graduation rates, may tell us just as much about the existing student population as it does about the selection process when applying.

Another insight is about university type. Many students contemplate private versus public universities, and many question private, for-profit universities. Selectivity is often associated with prestige, and although this is not always the case, it is something to be considered in the application process. For profit universities have a median of 90% acceptance rates, which may be related to their profit motives.

Tuition Dashboard

